National University
Of Computer and Emerging Sciences

Foundation of Advancement
Of Science and Technology

Lab Task:	10
Course Code:	CL2005
Course Title:	Database Systems - Lab
Semester:	Spring 2026
Instructor:	Muhammad Mehdi
Email:	muhammad.mehdi@nu.edu.pk

Guidelines

- Please review the lab manual before starting the task to ensure you fully understand the requirements.
- You may use **only the concepts covered in the lab manual** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
  Example: *24p-1234_ozair_labtask10.pdf*
- Unethical use of AI tools will result in a **deduction of 5 marks**.
- Late submissions will incur a penalty of **-1 mark per day**.

Tasks

You are provided with a SQL script file named `lab-10-llms.sql`, which contains SQL statements to create and populate the required tables using a simplified dataset based on popular large language models and their usage logs.

Execute the given script file in MySQL server using the `source` command to load the dataset. After successfully setting up the database, perform the following SQL queries:

1. Create a view named `active_models` that displays only active models along with their provider names and base prices.
2. Create a view named `usage_summary` that shows model name, total requests count, and total cost aggregated from usage_logs.
3. Using the `usage_summary` view, display models whose total cost exceeds the average total cost across all models.
4. Display user names concatenated with model names in the format: **"Ali used ChatGPT-4"** using `CONCAT`.

5. Display model names along with:

- length of the name
- last 5 characters of the name (use negative indexing)

6. Display cost as a string value using `CAST`.

7. Display cost, and replace `NULL` or zero cost values with **"Free Usage"** using `COALESCE`.

8. Display request_time formatted as: **"10-Jan-2024"** using `DATE_FORMAT`.

9. Convert the string **"15-02-2024"** into a date using `STR_TO_DATE`.

10. Display model names with their release dates formatted as **"Month Year"** (e.g., March 2023).

11. Show usage logs with request date formatted as **"Day, DD Month YYYY"**.

12. Display request_time and add 1 week to it using `DATE_ADD`.

13. Display request_time and subtract 1 month using `DATE_SUB`.

14. Calculate the difference in days between model release date and request_time using `DATEDIFF`.

15. Calculate:

- Response delay using `TIMEDIFF`
- Time taken in seconds using `TIMESTAMPDIFF`